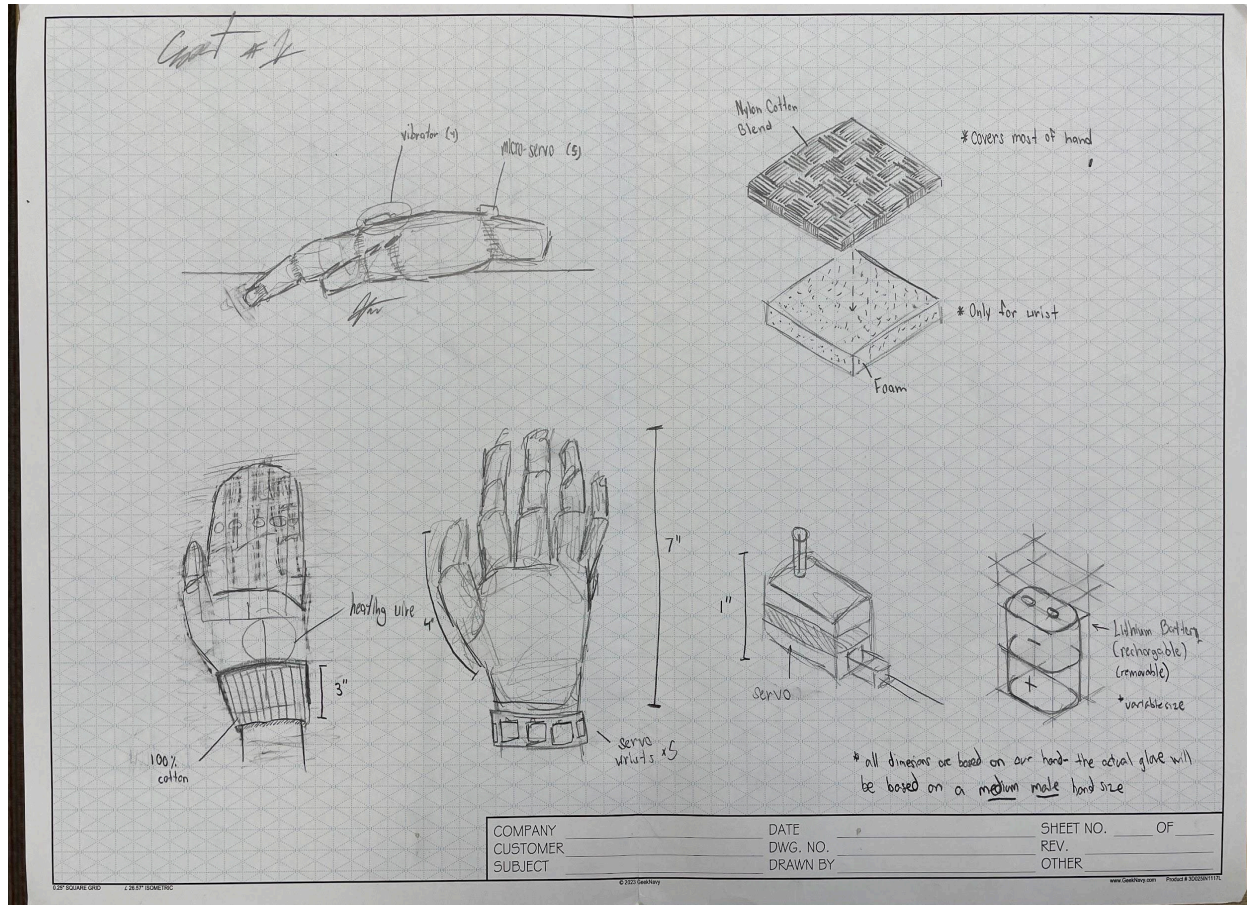


## Element D



### Potential Solution #1: The Handler

Our first solution, shown above, is called The Handler. The motherboard is an ESP32, which has Arduino and Bluetooth capabilities, so we can connect an app to it and also code it through the Arduino IDE. The heating is conducted through a heating wire, which is wrapped around all the fingers and connected to the motherboard, which sits on top of the wrist. The massaging is done through 4 micro-vibrators, 2 of which sit on the fleshy section of the thumb and 2 which sit in between the knuckles of the ring, middle, and index fingers. Grip is assisted through a fishing line that runs through the palm to each finger, and is kept in place by small loops, so when the hand contracts, the wire stays put. These wires are controlled by micro servos that are positioned on the forearm, on the opposite side of the ESP. The glove itself is made of a spandex cotton blend, and the cuff, which will hold the ESP and servos, is made of 100% cotton. For easier manufacturing, cheap cloth or plastic gloves can be made for the prototype. The lithium battery is on the wrist, next to the motherboard, and can be removed for charging, which allows for multiple batteries to be used in succession. The flex sensors are positioned

along where the heating wire is, but kept separate so the heat doesn't compromise or interfere with the sensors.

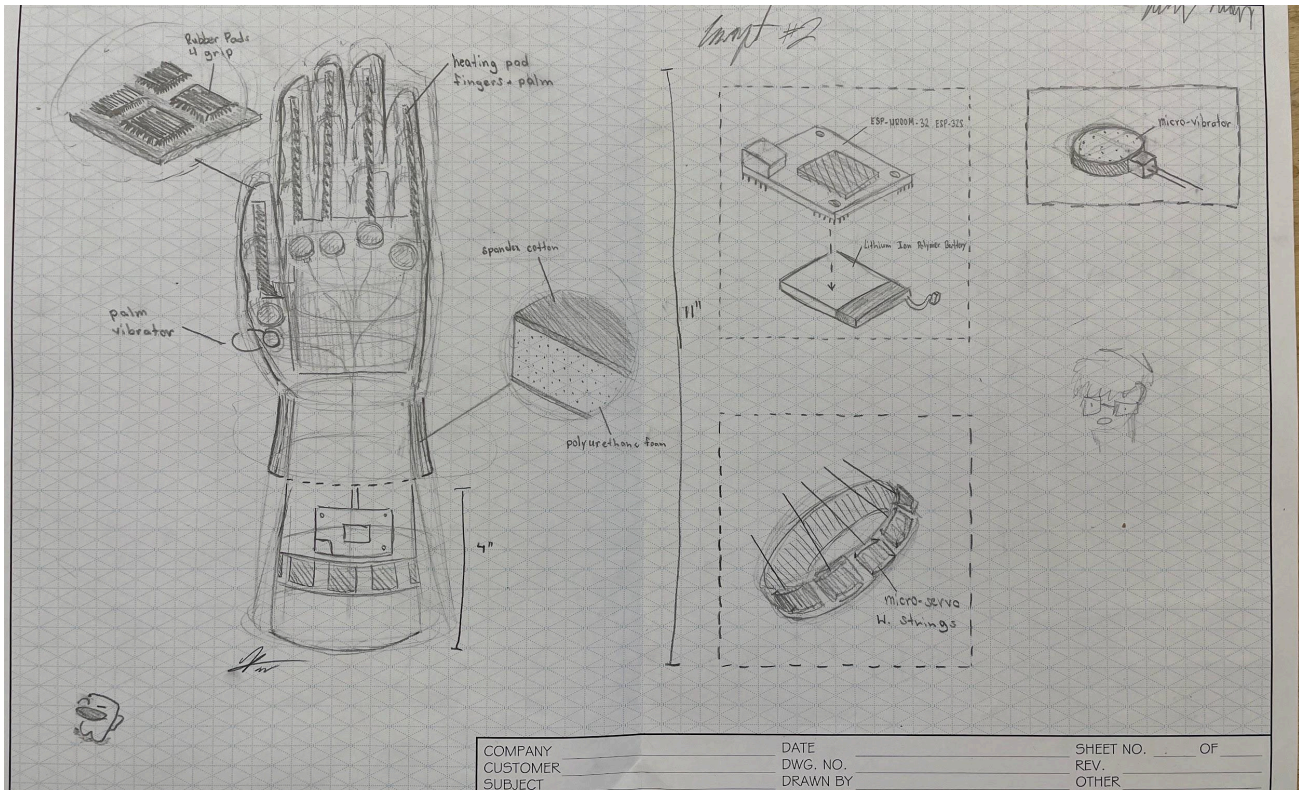
### **Benefits**

- Taking out the battery allows a user to use the glove for longer periods of time, since they can use multiple batteries.
- The heating wire is thin and simple to integrate into the glove material, and is also inexpensive.
- The fishing line is very strong and can be integrated into the servo to assist the gripper.
- There are Bluetooth capabilities, so an app can be used so the user can easily change settings.

### **Tradeoffs**

- It may be a challenge to make the battery easily accessible and removable from the rest of the device.
- The heating wire could cut into the fabric and wear it down easily, and it can also constrict the hand from closing all the way.
- The fishing line can also cut into and wear down the fabric, and it may be too slippery for the servos to handle.
- The battery life of the ESP may not be very high, since a lot of components are constantly being used.





## Potential Solution #2: Grippier

For our second design, Grippier, we kept some aspects constant from The Handler, like the ESP 32 motherboard. The heating is done through a heating pad charged via the LiPo battery and controlled with the ESP32. The heating pad is woven into the glove on the palm of the hand. There are 5 micro-massaging vibration motors, each behind each knuckle on your hand. These motors will enable massaging sessions for the hand that can be adjusted via the Bluetooth app. The massaging motors are controlled via the ESP32 and powered via the LiPo battery. There is 1 servo motor at the thick wrist section of the hand. The servo has 5 wires that are looped through guidance loops in the fingers to pull and release your fingers in order to bend them to enhance and assist with gripping objects. Each servo is controlled with a servo controller in the ESP32 and powered with the LiPo battery. Flex sensors will exist on the top of each finger, in order to measure the bending of your fingers to calculate the amount of force needed to assist with gripping and releasing grip when trying to grasp objects. On the fingers are rubber pads to help with grasping objects. The glove is made of a spandex cotton blend. These can be bought on Amazon or knitted, whichever is easier. On the wrist, the cuff will be thicker with polyurethane foam in between the fabric; this will be made by just cutting foam and wrapping it around the wrist. The wrist is thicker to provide wrist support to ease pain and discomfort in the hands. The ESP 32 will lie on the

wrist, and the rechargeable LiPo battery will be next to the microcontroller. The glove must be recharged periodically after the battery dies.

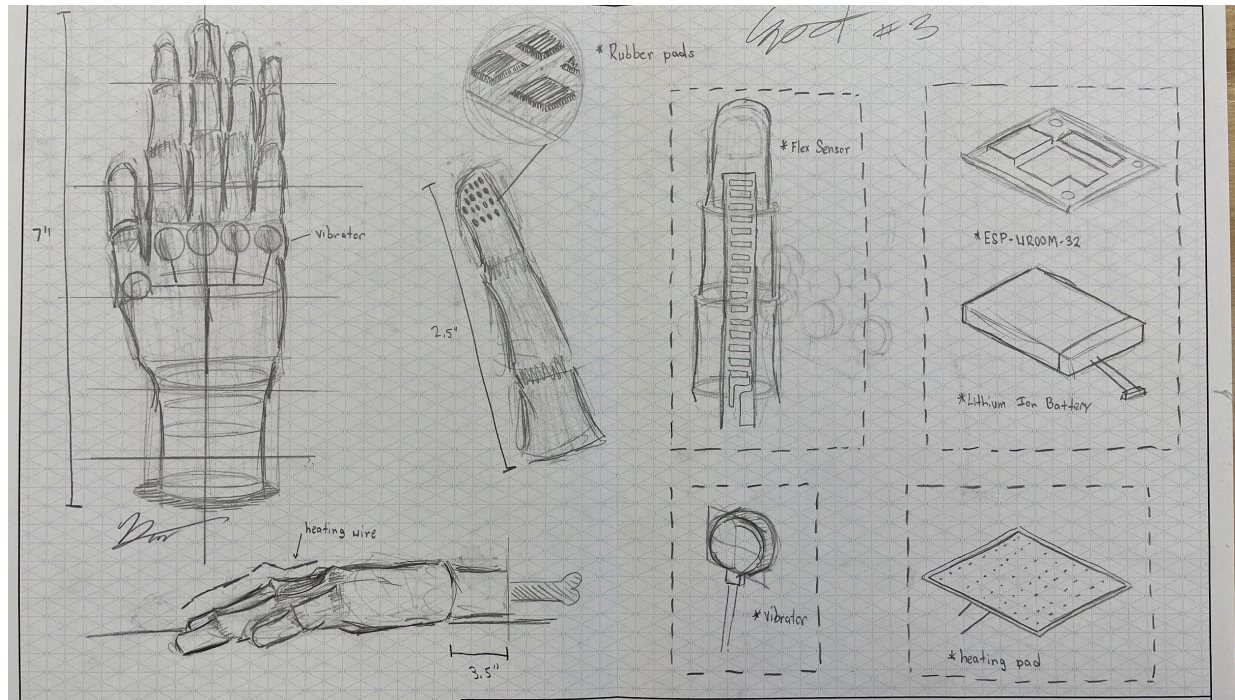
### **Benefits**

- The glove can be controlled via Bluetooth
- The fishing line being used is very strong and durable
- Using servos for each finger when assisting grip makes sure that the grip assistance is adapted to the differences in each finger's movement
- Guidance loops make sure that the finger movement is more natural
- Rubber pads help with holding small things
- A thick wrist with foam in between the fabric helps stabilise and support the wrist, and mitigates the loss of stability from the weight of the glove components.
- The micro-massaging motors help ease discomfort and stiffness in the hands
- The large heating pad on the palm helps ease pain and stiffness

### **Tradeoffs**

- Since you can't remove the battery and replace it immediately, battery life is limited to one charge before having to wait till it charges again.
- A bulkier wrist section means that the glove is less breathable and may get sweaty.
- A large heating pad on the palm leaves out other parts of the hand, like the fingers. When heating, it makes the glove a little bulkier.
- Small guidance loops might be difficult to manufacture
- Micro servos added to the wrist section add to the thick wrist section, making it increasingly bulky, makes the glove heavier





### Potential Solution #3: Artie (short for articulation)

Finally, our third design, Artie, combines many aspects of our first two designs. The motherboard continues to be an ESP32, and it uses a spandex cotton blend, with a 100% cotton wrist for full support. It is powered by a LiPo battery that is attached to the glove and cannot be removed. To charge, the glove must be taken off and a USB plugged in directly from an outlet. The flex sensor is on the index and thumb, and is attached lengthwise to the glove as shown. There will be rubber pads on the tips of the fingers for better grip, and the grip will be assisted by fishing line held by servos, looped through the fingers. There will be 5 vibration motors on top of the hand, one for each knuckle, and a heating pad on the palm, as well as heating wire through each of the fingers. There are 5 servo motors at the thick wrist section of the hand. Each servo has a wire that is looped through guidance loops in the fingers to pull and release your fingers in order to bend them to enhance and assist with gripping objects. Each servo is controlled with a servo controller in the ESP32 and powered with the LiPo battery.

### Benefits

- Heating-based—this design can provide sufficient and even heat to the hand
- Sufficient amount of vibration motors to ensure proper massage
- Rubber pads help grip objects more easily
- 5 vibration motors ensure even massaging
- A cotton cuff will help the wrist bear the weight

## Tradeoffs

- The glove will be heavy because of the 5 motors and 5 servos
- The glove cannot be worn while charging
- A flex sensor may not perceive changes in fingers other than the index and thumb.
- The string may rub against fingers, causing irritation

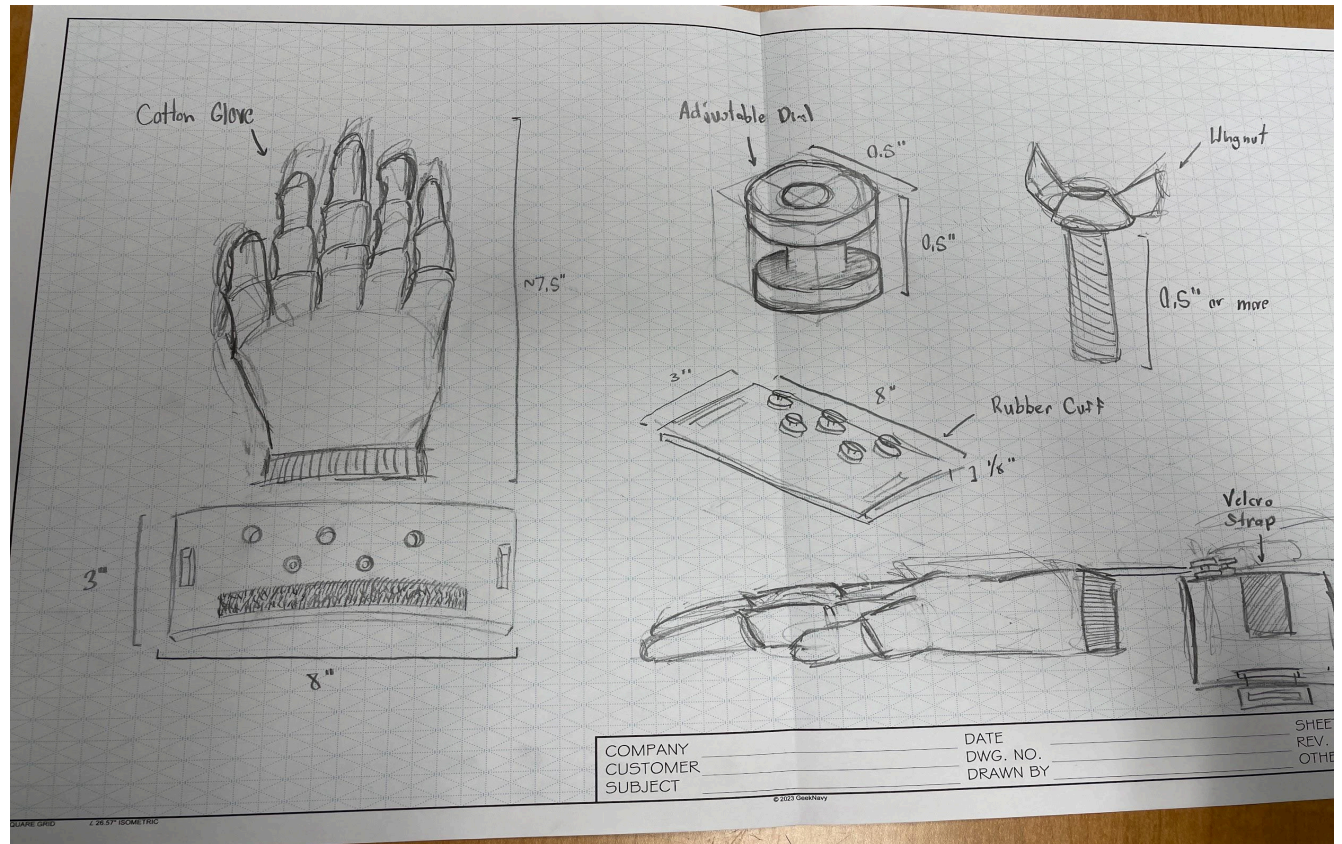
## Final Choice

| Design Reqs                                                                                                                                                                                                                                                                                                             | Weight      | The Handler (#1) |             | Grippier (#2) |             | Artie (#3) |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------|-------------|---------------|-------------|------------|-------------|
|                                                                                                                                                                                                                                                                                                                         |             | Rating           | Value       | Rating        | Value       | Rating     | Value       |
| The glove will ensure customer safety by regulating temperature, including failsafes to mitigate abnormal temperature fluctuations, and protect from voltage surges by a measure of internal pass/fail tests, after running the product at maximum intensity.                                                           | 20%         | 4                | 0.80        | 4             | 0.80        | 4          | 0.80        |
| The glove will assist grip, measured by stakeholder pain relief and ease in grip using a rating scale of 1 - 10 (1 being the worst), measured before and after using the glove, intending for a difference of 3 or greater. General objects that patients have difficulty holding/using will be identified for testing. | 20%         | 4                | 0.80        | 3.5           | 0.70        | 4          | 0.80        |
| The glove will heat the hand(s) up evenly to a temperature of 104 degrees Fahrenheit, not exceeding 111 degrees Fahrenheit, measured in degrees Fahrenheit using a built-in thermometer, heating for 15-20 minutes.                                                                                                     | 15%         | 3                | 0.45        | 3             | 0.45        | 3.5        | 0.53        |
| Built-in glove sensors will sense/perceive changes in temperature, grip setting, movement, and vibration in the glove worn on the hand afflicted by arthritis, measured using data from built-in sensors, including, but not limited to flex sensors, thermometers, timers, etc.                                        | 15%         | 4                | 0.60        | 4             | 0.60        | 4          | 0.60        |
| The glove shall easily release the grip for people with arthritis measured with a consumer rating of at least a score of 7 or greater, measured by stakeholder feedback on a scale of 1 to 10, 1 being the worst.                                                                                                       | 9%          | 4                | 0.36        | 4             | 0.36        | 4          | 0.36        |
| The glove will manipulate the soft tissue in the hands for people with hand arthritis for 15 minute sessions with a built in timer to measure the duration for each session.                                                                                                                                            | 14%         | 3.5              | 0.49        | 3             | 0.42        | 4          | 0.56        |
| The glove will adapt to basic app settings for people with hand arthritis by continuously updating to any recent changes made on the connected app via bluetooth by a measure of consumer rating/feedback on a scale of 1 - 10, with 1 being the worst.                                                                 | 7%          | 4                | 0.28        | 4             | 0.28        | 4          | 0.28        |
| <b>Total</b>                                                                                                                                                                                                                                                                                                            | <b>100%</b> |                  | <b>3.50</b> |               | <b>3.33</b> |            | <b>3.65</b> |

Our decision matrix can be seen here with our design requirements from element C, along with a special weighting we calculated based on the importance of each requirement. Based on the matrix alone, our third design, Artie, did the best, with a score of 3.65. Safety and assistance are rated the highest because in medical technology, safety is the most important, and the assistance of grip is the main function of the glove. The massaging, heating, and sensing are weighted less, but are still important because they are secondary functions of the glove. App settings and grip release are weighted less because the app isn't the most important to get done, and grip release requires minimal work due to most of the grip function being covered in the assist grip requirement. However, after receiving feedback from multiple experts, including Drexel professors, a hand therapist, and our teacher, we have slightly amended our plan of action and also our design requirements from Element C. Essentially (and this will be further explained in the plan of action), we will take the glove creation step by step, and add on each feature if we have time.



## Final Design:



This design is the final decision and reflects the final prototype design. It has taken all the best aspects of the previous conceptual designs and taken into consideration the exclusion of certain properties and components, such as the heating elements, vibration elements, and Arduino elements. The design is composed of a single glove, which can be whatever size fits the customer as needed, and a wrist band made of flexible material (rubber). The glove will be stitched through with a line per finger, covering the face of the hand and then alternating to the palm side of the hand to the tips of the fingers, where it is stitched. This is so that the applied tension to the string forcibly bends the individual fingers. The band holds five adjustable knobs that hold the string; they can be adjusted to account for the various lengths of differing finger and hand types. Attached to the band is a velcro strap in order to provide the string a stable anchor to ensure each string is taut.

## Benefits:

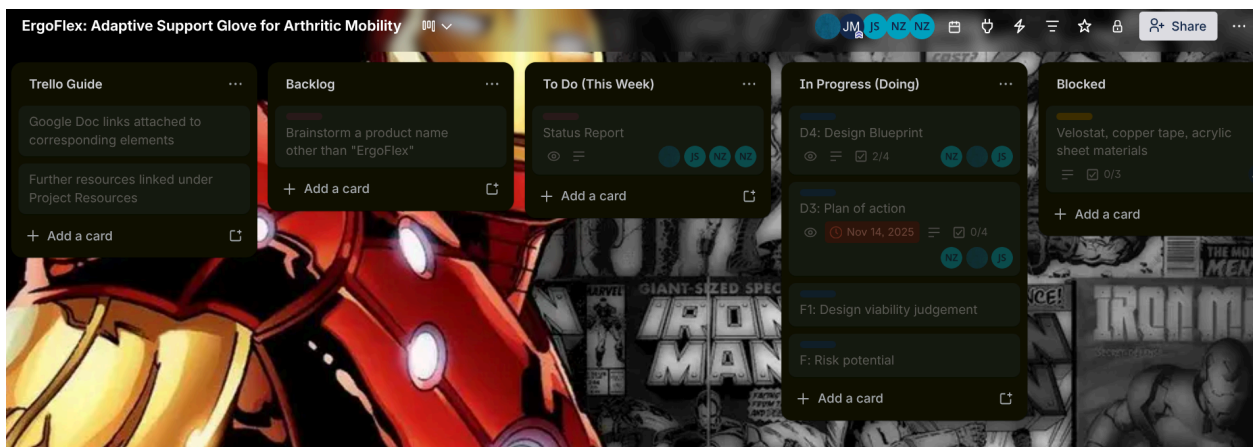
- Adjustable through the knobs
- Simple to manufacture/create a prototype of because of fewer parts

- No need for a battery or an electrical system
- The Velcro strap allows for easy adjustability and taking it on/off
- The simple mechanism can allow for repeatability

### Tradeoffs:

- Bulkier; certain everyday tasks can be prohibited or restricted in sorts
- Reduction in effectiveness against arthritis
- Can be less durable as the fabric stretches, and the string could snap
- A rubber cuff is not as comfortable as a soft cotton one, and smells more
- Harder to adjust because the wing nuts must be operated by someone else

### Plan of Action



*(Screenshot of Trello workspace for planning tasks)*

A day-by-day timeline has been set up via Trello, where all planned actions are created in tasks and properly recorded with a description of context and goal. For more intricate details, visit Trello [here](#).

A brief overview of the planned production goes like this: Through the created Bill of Materials, acquire all necessary materials and equipment. Those are sewing string, a printed glove pattern, Nylon or Cotton-based fabric, Velostat plastic, copper foil, LiPo batteries, DC motors, and an ESP32. The first process in the design is to construct a basic glove made out of the chosen material, Nylon or Cotton (or both). Print out the pattern (our group acquired it from [here](#)), then cut out the pattern from fabric with an excess sewing margin. Finally, sew the parts together, and you will get a basic fitting glove. Next is to test and construct the measuring system of the Velostat plastic with copper foil to the ESP32, also connected to the LiPo batteries. These plastic copper foils will be attached to the fingers, and trail towards the ESP32 located on the wrist, where the allocated DC motors will move based on the responsiveness of the Velostat to assist in the hand movements. After basic testing and a primitive prototype has been made, our



group will come together to brainstorm developments for an improved design and various changes.

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### **Project Update: 11/20/2025**

The current prototype decision has been developed into a different design. For the time being, we will focus prototype development on the grip assisting aspect. Reviewing previous feedback received recommended that the scope of the project be dialled back to a realistic goal or separate aspects to focus on them separately.

The current prototype concept features passive assistive grip technology by utilising the constant tension of a string to manipulate finger positioning on command, completely by the user.

#### **Benefits of this Decision:**

- Simplification of technology
- Low-cost to produce and replicate
- Easier to build on (technology-wise)
- Larger timeframe

#### **[Amended] Plan of Action:**

The day-by-day timeline for the project has been modified to reflect the change in the project by temporarily placing initial priorities like heating and soft-muscle manipulation into the “backlogged” category. For now, a passive grip assistive mechanism has been developed in the form of non-flexible strings. By taking advantage of the constant length of the string and its inability to flex, we can forcibly manipulate the fingers by movement of the wrist, and therefore reduce the strain of arthritis of the hand joints, which meets the initial priority of the project. The Bill of Materials will remain as it is, with a few additional products such as silicon foam pads and flexible sewing string to meet the newly developed mechanism. An additional glove will be constructed and sewn along with the development of a 3D [BOA-Dial](#) to adjust the tautness of the string per user.

#### **Update:**

We have excluded the use of a BOA-Dial system due to the complexity of implementing it at the micro-scale. Available 3D prints account for one thread, and the purpose intended was for the dial to account for all 5 threads to be adjustable. Therefore, the component

cannot be implemented. Everything remains as it is, and a system of five adjustable knobs (held in place with wing nuts) is to be implemented instead.